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Making Old Film Great Again: Degradation-aware State Space Model for Old Film Restoration

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Abstract

Unlike modern native digital videos, the restoration of old films requires addressing specific degradations inherent to analog sources. However, existing specialized methods still fall short compared to general video restoration techniques. In this work, we propose a new baseline to re-examine the challenges in old film restoration. First, we develop an improved Mamba-based framework, dubbed MambaOFR, which can dynamically adjust the degradation removal patterns by generating degradation-aware prompts to tackle the complex and composite degradations present in old films. Second, we introduce a flow-guided mask deformable alignment module to mitigate the propagation of structured defect features in the temporal domain. Third, we introduce the first benchmark dataset that includes both synthetic and real-world old film clips. Extensive experiments show that the proposed method achieves state-of-the-art performance, outperforming existing advanced approaches in old film restoration. The implementation and model is available at https://github.com/MaoAYD/MambaOFR.

1. Introduction

The preservation and restoration of old films are crucial for safeguarding the cultural and historical heritage encapsulated within the moving images of the past. As physical film materials deteriorate over time, the rich narratives and artistic expressions they convey risk being lost forever. Digitization offers the possibility of revitalizing old films [51]. However, due to limitations in the technology available at the time of filming, as well as factors such as preservation conditions, old films often lack the visual appeal necessary for modern audiences This limits their dissemination on modern media platforms, such as streaming services. Therefore, developing automated methods for the restoration of old films [19, 47] to replace labor intensive manual work of restoration professionals is of significant importance for the preservation of old films.

Removing degradation from old films is a complex issue. There are two primary challenges. First, old films suffer from complex degradation under both analog and digital conditions, resulting in mixed and often unknown types of deterioration. This complexity poses challenges for various video restoration methods that typically target single types of degradation [8, 11, 40]. Second, old films typically exhibit structural defects, such as cracks, corrosion spots, and stripe noise, which are prominent and significantly compromise the internal integrity of video frames. These degradations are difficult to automatically distinguish from the true content of the frames and are challenging to remove using adjacent pixel information within a single frame.

Current methods remain significantly limited in addressing these issues. Some video restoration techniques [6, 28, 31] attempt to tackle complex mixed degradation within a single framework, but these methods primarily focus on unstructured degradations such as noise and blur, which struggle to comprehensively address the broader challenges of old film restoration. In contrast, methods specifically designed for old film restoration [46, 47] attempt to explicitly address structured defects. However, these approaches fail to effectively leverage complementary temporal features or aggregate global spatial features. Furthermore, they have not comprehensively considered the propagation of structured defects across the temporal domain. As a result, these methods struggle to achieve restoration results on par with state-of-the-art video restoration techniques.

In this work, we propose a new baseline to re-examine the challenges of old film restoration. Our goal is to adaptively address all types of degradation in old films within a unified framework. We develop an enhanced Mamba-based [15] framework, dubbed MambaOFR, for old film restoration. Taking advantage of the long-range modeling capability of 2-D selective state space [16] and an efficient global receptive field, our framework effectively aggregates spatial complementary features for restoration. Building

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on this, we propose an adaptive degradation-aware removal approach by integrating the proposed degradation-aware prompt block to handle multiple complex degradation conditions present in old films. In the temporal domain, we observe that some difficult-to-remove structured defects propagate between frames during temporal feature aggregation. To address this, we propose a flow-guided mask deformable alignment module for feature alignment across the temporal sequence. In addition, due to the long-standing absence of a unified public dataset and benchmark in the field of old film restoration, we collect and propose the first benchmark dataset that includes both synthetic and real-world old film clips. Our contributions can be summarized as follows:

- We propose a recurrent old film restoration framework, MambaOFR, which dynamically adjusts degradation removal patterns by generating degradation-aware prompts to address the complex and composite degradations present in old films. Moreover, to mitigate the propagation of structured defect features in the temporal domain, we propose a flow-guided mask deformable alignment that mitigates the propagation of structured defect features in the temporal domain.
- Due to the lack of a public benchmark for old film restoration, we collect and propose the first large-scale old film restoration dataset, which consists of 30 synthetic video clips and 216 real-world old video clips with diverse scenes.
- Extensive experiments on the proposed dataset demonstrate that the proposed method achieves state-of-the-art performance in old film restoration, outperforming existing advanced old film restoration and video restoration methods.

2. Related Work

2.1. Video Restoration

The objective of video restoration is to mitigate the degradation of video content to produce high-quality video frames. Existing methodologies attempt to establish a correspondence between the original video and its degradation to accurately model the restoration process, such as video denoising [12, 43], video deblurring [56, 58], and video super-resolution [21, 61]. Several restoration tasks involve the development of intricate physical models to accurately represent real-world degradation processes [42, 53, 55]. The primary challenge in video restoration lies in rehabilitating degraded frames from temporal and spatial features. Recent advances in neural network foundational models [10, 36, 63] have significantly improved the ability of models to extract and aggregate spatial features, resulting in substantial performance improvements [16, 27, 33, 52, 57]. These advances have also been applied to address spatial degradation in videos [3, 35]. However, unlike image restoration, leveraging the inter-frame dependencies between adjacent video frames is crucial for effective video restoration. Usually, optical flow estimation is employed to predict pixel motion between video frames and align features by warping [18, 54]. To overcome the limitations of optical flow, deformable convolution [7] has been introduced to address feature alignment in challenging scenarios [3, 45, 49]. Meanwhile, the attention mechanism is also used to handle the temporal dependence in implicit [2, 32]. Nonetheless, these methods fail to account for the uncertainty in optical flow estimation caused by occlusions resulting from specific degradations in old films. Moreover, they do not consider the propagation of such degradations in the temporal domain, thereby not achieving satisfactory performance in old film restoration.

2.2. All-in-One Restoration

Models designed for single degradation types are inadequate for addressing restoration tasks involving complex degradations. Some efforts [28, 31, 59] attempt to combat multiple degradations with a single model. These All-In-One image restoration methods aim to learn some specific latent spaces to disentangle the content from degradation. Li *et al*. [31] built a generative network with multiple specific degradation encoders to handle multiple degradations in a single network. However, these methods require prior knowledge of the type of degradation to select the appropriate encoder, which hinders the end-to-end implementation of All-In-One models. Li *et al*. [28] introduced contrastive learning to learn a unified degradation feature extractor to provide degradation hints for the restoration network. Potlapalli *et al*. [41] proposed a degradation prompt generation module, which implicitly learns the type of degradation, can dynamically generate the degradation prompt and adaptively adjust the network parameters. To explore a unified latent space that represents images in high-dimensional space for clean image reconstruction, Zheng *et al*. [62] proposed to learn a diffusion model with a shared degradation distribution. Cheng *et al*. [6] uses a sequence-wise adaptive degradation estimator to estimate corrupted video and proposes an inter-frame fusion strategy to decrease the propagation of temporal degradation. Unlike previous works, the old film restoration must contend with indeterminate degradation types and composite degradation, making explicit modeling of the degradation challenging. We propose a degradation-aware dynamic parameter convolution module to model the current degradation implicitly. This approach enhances the degradation dynamic adaptation capability of the mamba model at a cost of only a small number of parameters and computational complexity.

2.3. Old Film Restoration

Conventionally, traditional methods [13, 17, 25] divide old film restoration into two stages, utilizing contaminant (i.e. scratches, dirt, or dust) detection as a preliminary step and hints for the subsequent restoration process. However, hand-crafted features limit the accuracy of contaminant detection, and the restoration task is treated as an inpainting process, thereby neglecting low-level distortions, which results in the final visual quality falling short of satisfactory levels. Subsequently, Iizuka *et al*. [20] introduced a fully 3D convolutional method designed to restore old films. Although 3D convolution achieves notable performance improvements over traditional methods, it struggles to model complex temporal relationships. Additionally, the lack of a global receptive field hinders the utilization of globally relevant features. Wan *et al*. [47] proposed a recurrent transformer network and implicitly modeled contaminant features. The significant computational burden associated with transformer-based architectures makes the computation under a global receptive field impractically heavy. Lightweight designs based on window attention [36] typically come at the expense of the global receptive field. The trade-off dilemma between efficient computation and global modeling remains fundamentally unresolved. Our proposed MambaOFR model features a global receptive field with low computational complexity, while concurrently accounting for dynamic degradation adaptation. This advancement has elevated the performance of old film restoration tasks to a new level.

3. Method

3.1. Preliminary

Recently, a class of sequence models represented by structured state-space models (S4) has been highly affected by continuous linear time-invariant systems. The S4 models map a one-dimensional sequence $x(t) \in \mathbb{R} \rightarrow y(t) \in \mathbb{R}$ through an N-dimensional hidden state $h(t) \in \mathbb{R}^N$, which can be formulated as a linear ordinary differential equation:

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t), \\ y(t) &= \mathbf{C}h(t) + \mathbf{D}x(t), \end{aligned} \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$, $\mathbf{B} \in \mathbb{R}^{N\times 1}$, $\mathbf{C} \in \mathbb{R}^{N\times 1}$ and $\mathbf{D} \in \mathbb{R}$. To facilitate the optimization of deep learning methods, the continuous system described by Equation (1) is discretized using the zero-order hold (ZOH) rule, which can be formulated as:

$$\begin{aligned} \bar{\mathbf{A}} &= \exp(\Delta\mathbf{A}), \\ \bar{\mathbf{B}} &= (\Delta\mathbf{A})^{-1}(\exp(\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B}. \end{aligned} \tag{2}$$

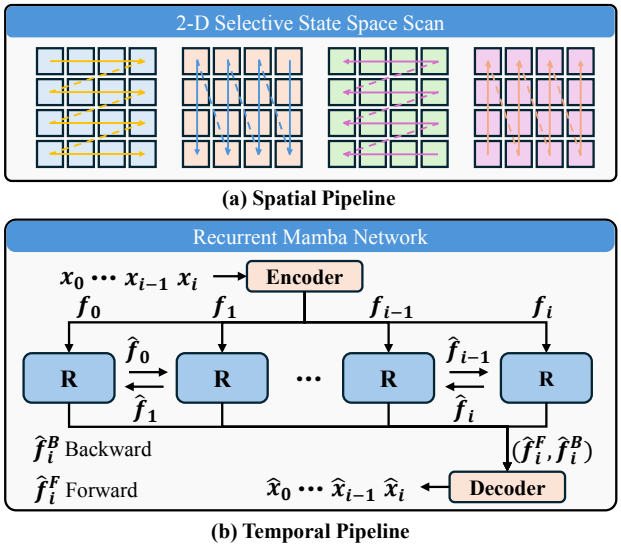


Figure 1. The main flow of the proposed framework in the spatial and temporal pipeline. The top row shows the four-direction scan of the selective state space module. The bottom row shows the forward-backward recurrent computation of a frame sequence. **R** is the restoration network.

The RNN form of Equation (2) with step size Δ can be rewritten as:

$$\begin{aligned} h_k &= \bar{\mathbf{A}}h_{k-1} + \bar{\mathbf{B}}x_k, \\ y_k &= \mathbf{C}h_k + \mathbf{D}x_k. \end{aligned} \tag{3}$$

To facilitate parallel operator optimization, the Equation (3) can similarly be written in CNN form:

$$\begin{aligned} \bar{\mathbf{K}} &\triangleq \left(\mathbf{C}\bar{\mathbf{B}}, \mathbf{C}\bar{\mathbf{A}}\bar{\mathbf{B}}, \cdots, \mathbf{C}\bar{\mathbf{A}}^{L-1}\bar{\mathbf{B}}\right), \\ \mathbf{y} &= \mathbf{x} \circledast \bar{\mathbf{K}}, \end{aligned} \tag{4}$$

where $\bar{\mathbf{K}} \in \mathbb{R}^L$ denotes the convolution kernel for L length sequence inputs while $\circledast$ is the convolution operation conducted to input sequence $\mathbf{x}$. Mamba [15] model further improves the State Space model to a Selective State Space with input-dependent $\mathbf{A}, \mathbf{B}, \mathbf{C}, \Delta$ matrices. This efficient context-aware schema can be easily extended to 2-D data.

As illustrated in Figure 1, in the spatial pipeline, the 2-D Selective Scan Module (2D-SSM) enables the network to preserve the local spatial relation of image patches while aggregating global features by scanning image patches from four different directions. In the temporal pipeline, we use frame-level moving windows to construct a recurrent feature aggregation network to extract complementary information between frames to reconstruct the clean frames.

3.2. Network Architecture

Figure 2 illustrates the proposed MambaOFR for old film restoration. Specifically, the pipeline of our framework

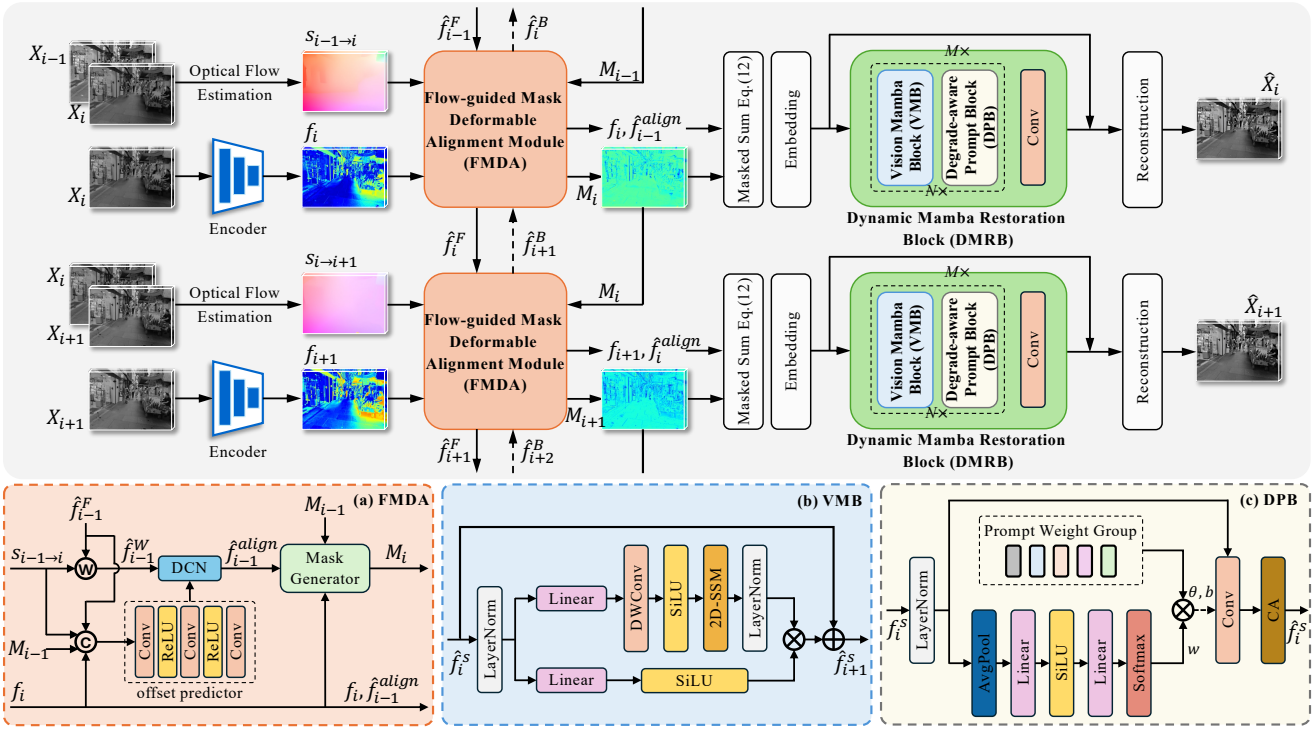


Figure 2. Overview of proposed MambaOFR. The $\boldsymbol{W}(\cdot)$ denotes the warping operation and $\boldsymbol{C}(\cdot)$ denotes the concatenation operation. The upper part of the figure shows the general flow of recurrent computation of the proposed method.

is divided into two main parts: temporal domain feature alignment and spatial domain feature restoration. Given a sequence of degraded old film frames with T length $\{\boldsymbol{x}_i \in \mathbb{R}^{3\times H\times W} | i \in [0,T)\}$, we use an encoder to extract the frames to shallow features $\{\boldsymbol{f}_i \in \mathbb{R}^{C\times H/4\times W/4} | i \in [0,T)\}$, where C, H, and W are the channel, height, and width, respectively.

For the first frame, the feature $\boldsymbol{f}_0$ will be directly fed into the spatial domain feature restoration network. After embedding, the Vision Mamba Block (VMB) learns to aggregate the related feature by four-direction sequence scanning, while the Degradation-aware Prompt Block (DPB Sec. 3.3) extracts global context to generate degradation prompts to adaptively adjust network parameters to eliminate feature degradation.

Then, the previously restored features $\hat{\boldsymbol{f}}_0$ will be input into the temporal feature alignment module (Sec. 3.4) simultaneously with the features of the next frame $\hat{\boldsymbol{f}}_1$ to generate a mask for removing specific degradations from old films and perform optical flow-based feature alignment. The aligned feature will be fed to the restoration network for the next step. The whole process will undergo a forward-backward computation in the time domain. As illustrated in Figure 1 (b), we finally reconstruct the final clean frames $\{\hat{\boldsymbol{x}}_i \in \mathbb{R}^{3\times H\times W} | i \in [0,T)\}$ by combining the forward-backward restored features $\{(\hat{\boldsymbol{f}}_i^F, \hat{\boldsymbol{f}}_i^B) | i \in [0,T)\}$.

3.3. Degradation-aware Prompt Block

In old films, degradation rarely occurs in isolation. Constrained by the technological limitations of preservation methods, filming equipment, and the quality of film digitization, old movies typically exhibit multiple types of degradation. Consequently, it is crucial for networks to possess the capability to adapt to unknown degradations. Prompt-based approaches have demonstrated remarkable performance in visual tasks [22, 24], particularly in restoration tasks [29, 41]. This paradigm is inherently suitable for injecting explicit adaptive dynamic parameters into the restoration network. However, unlike the transformer architecture, the interaction sequence between adjacent blocks in the Mamba architecture is constrained by the scanning direction. The selective memory mechanism makes direct prompt injection in 2D-SSM operations difficult to produce a global interaction. However, the convolution layer after the SSM operation plays an important role in recovering the neighborhood similarity of the features [16].

Hence, we propose the Degradation-aware Prompt Block to inject the prompt. Inspired by dynamic convolution [5], we init a group of convolution kernels $\{\boldsymbol{\theta}_i \in \mathbb{R}^{C\times 3\times 3} | i \in [0,N)\}$ and biases $\{\boldsymbol{b}_i \in \mathbb{R}^{C\times 1} | i \in [0,N)\}$ as a prompt. As illustrated in Figure 2 (c), given a deep feature $\boldsymbol{f}_i^S \in \mathbb{R}^{C\times H\times W}$ from the previous VMB, we first normalize the feature with layer normalization. Then, the feature will be

fed to a stack of linear layers $FC(\cdot)$ after the global average pooling $AP(\cdot)$ to generate the prompt weights $\boldsymbol{\omega} \in \mathbb{R}^{1\times N}$, which can be formulated as:

$$\boldsymbol{\omega} = \sigma(FC(SiLU(FC(AP(\boldsymbol{f}_i^S))))), \tag{5}$$

where $\sigma(\cdot)$ and $SiLU(\cdot)$ are the Sigmoid and SiLU functions respectively. The dynamic convolution parameters can be derived from the following formulas:

$$\tilde{\boldsymbol{\theta}}(x) = \sum_{i=1}^{N} \boldsymbol{\omega}_i(x)\boldsymbol{\theta}_i, \quad \tilde{\boldsymbol{b}}(x) = \sum_{i=1}^{N} \boldsymbol{\omega}_i(x)\tilde{b}_k, \tag{6}$$

where x denotes the input features with various degradations. The dynamic parameters $\tilde{\boldsymbol{\theta}}, \tilde{\boldsymbol{b}}$ will be employed in the next convolution layer to generate the prompt features $\boldsymbol{f}_i^P$. We aggregate it with the original deep feature $\boldsymbol{f}_i^S$. The processes are as follows:

$$\begin{aligned} \boldsymbol{f}_i^P &= Conv(\boldsymbol{f}_i^S|\tilde{\boldsymbol{\theta}}, \tilde{\boldsymbol{b}}), \\ \hat{\boldsymbol{f}}_{i+1}^S &= Attention(Cat(\boldsymbol{f}_i^P, \boldsymbol{f}_i^S)), \end{aligned} \tag{7}$$

where $Cat(\cdot)$ denotes the concatenate operation. The dynamically modulated features can carry degradation-related prompt information into the next VMB, endowing the network with the capability to adapt to degradations.

3.4. Flow-guided Mask Deformable Alignment

Unlike other video restoration tasks, old films possess a unique category of degradation (*e.g.,* scratches, banding noise, etc.). This type of noise tends to appear and disappear abruptly along the video timeline, resulting in more severe optical flow estimation errors. Although the previous work [47] has utilized explicitly learned soft masks to eliminate structural defects, the propagation of this degradation in the temporal domain and its impact on feature alignment have not been adequately considered.

To address this, we proposed the Flow-guided Mask Deformable Alignment to eliminate the defects. Given a restored feature of previous frame $\hat{\boldsymbol{f}}_{i-1}$, the module aims to align the previous feature $\hat{\boldsymbol{f}}_{i-1}$ to the current feature $\boldsymbol{f}_i$. We estimate the optical flow $\boldsymbol{S}_{i-1\to i}$ of the two frames using a pre-trained flow estimator [44]. The warped feature $\hat{\boldsymbol{f}}_{i-1}^W$ can be obtained as follows:

$$\hat{\boldsymbol{f}}_{i-1}^W = Warp(\hat{\boldsymbol{f}}_{i-1}, \boldsymbol{S}_{i-1\to i}), \tag{8}$$

where $Warp(\cdot)$ denotes the warping operation. Due to the optical flow error, the feature contains numerous mistaken alignments. We refine using deformable convolutions (DCN) as per standard practice [3]. As illustrated in Figure 2 (a), we specifically input the mask $\boldsymbol{M}_{i-1}$ from the previous stage, along with other features, into the computation of offsets, which contains explicit information about the location and shape of structural defects. The process is as follows:

$$\boldsymbol{O}_i = OP(Cat(\boldsymbol{S}_{i-1\to i}, \hat{\boldsymbol{f}}_{i-1}^W, \boldsymbol{M}_{i-1})), \tag{9}$$

where $\boldsymbol{O}_i$ is the offset of the deformable convolution layer generated by the offset predictor $OP(\cdot)$. Then, we can obtain the aligned feature by the deformable convolution as follows:

$$\hat{\boldsymbol{f}}_{i-1}^{align} = DCN(\hat{\boldsymbol{f}}_{i-1}^W|\boldsymbol{O}_i). \tag{10}$$

To prevent the temporal propagation of structural defect degradation, we incorporate the previous mask $\boldsymbol{M}_{i-1}$ into the prediction of the new mask as follows:

$$\boldsymbol{M}_i = MG(Cat(\hat{\boldsymbol{f}}_{i-1}^{align}, \boldsymbol{f}_i, \boldsymbol{M}_{i-1}))), \tag{11}$$

where $MG(\cdot)$ is the mask generator, which consists of several convolution layers and a Sigmoid function. Finally, we fuse these features with the new soft mask as follows:

$$\tilde{\boldsymbol{f}}_i = (1 - \boldsymbol{M}_i) \cdot \hat{\boldsymbol{f}}_{i-1}^{align} + \boldsymbol{M}_i \cdot \boldsymbol{f}_i. \tag{12}$$

The fusion feature will be fed to the restoration network for further processing.

3.5. Loss Function

To optimize our proposed method and restore visually pleasing results from old films, following the general setup in the restoration tasks, we employ the following loss functions.

$$\mathcal{L}_{\text{total}} = \lambda_1\mathcal{L}_1 + \lambda_p\mathcal{L}_{\text{P}} + \lambda_G\mathcal{L}_G. \tag{13}$$

Specifically, λ_1 is the L1 loss, λ_P is the perceptual loss [23], and λ_G is the spatial-temporal adversarial loss [4, 14]. The hyper-parameters $\lambda_1, \lambda_P, \lambda_G$ are empirically set to $\{1.0, 1.0, 0.01\}$. Detailed definitions are provided in the *supplementary material*.

4. Old Video Dataset

Since the lack of publicly available benchmark datasets, we construct a large-scale dataset to investigate the generalization and robustness of various old film restoration methods.

Data Synthesis To obtain a more precise evaluation using full-reference quality metrics, we synthesized old movie videos with ground truth from the validation set in the REDS dataset [39]. Following previous work [19, 46, 47], we employ compound degradation (including noise, blur, compression, etc.) as well as aging masks to simulate structural defects typical in old movies. Further details are provided in the *supplementary materials*.

Dataset Collection Given an old video from the internet, we can derive two observed findings: (1) due to the capturing time and saving conditions, old videos tend to be degraded by diverse frame artifacts, such as intensive noise

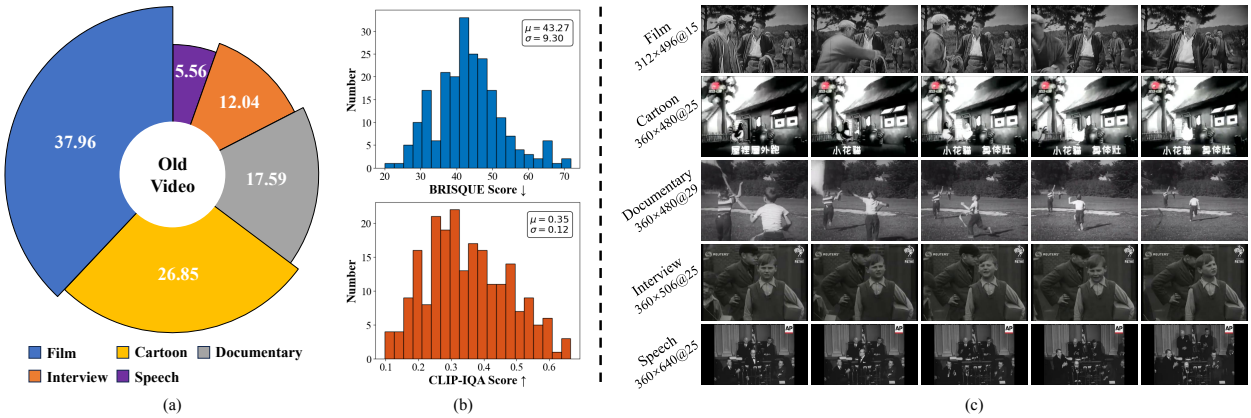


Figure 3. Visual examples and statistics of the proposed dataset. The data we collected primarily focuses on old films, while also encompassing a wide range of other scenes and variable spatiotemporal resolutions.

and motion blur; (2) different capturing devices in the past are prone to result in old videos along with different resolutions and frame rates, which pose a challenge for restoration methods. In order to improve the diversity of collected old videos, we first search the old videos with different categories and capturing time, and then carefully extract multiple 5-second clips from a video while ensuring low scene repetition. Finally, we obtain 216 old videos with 29,256 frames in total.

Dataset Statistics As illustrated in Figure 3 (a), we report the proportion of different categories of old videos we collected, including film, cartoon, documentary, interview as well as speech. To quantitatively analyze the distribution of the degraded frame quality, we introduce two no-reference metrics, *i.e.,* the BRISQUE [37] and CLIP-IQA [48]. Specifically, the BRISQUE adopts distortion-generic statistical features to evaluate the frame quality, while the CLIP-IQA performs the perception-generic assessment with the visual language prior CLIP. Figure 3 (b) illustrates the quality statistics of old video frames. It can be seen that the statistical differences between these two metrics indicate significant quality fluctuation between videos, which also means large restoration challenges. We further show some old video examples with diverse resolutions and frame rates in Figure 3 (c). More examples can be seen in the supplementary material.

5. Experiment

5.1. Experimental Setup

Implementation details The proposed MambaOFR is trained on two NVIDIA L40 GPUs using the PyTorch platform. We use the Adam optimizer ($\beta = (0.9, 0.999)$) [26] to optimize the model for a total of 200K iterations At each iteration, the input is a randomly selected 7-frame clip from the original video, which is randomly cropped to 256×256 pixels. For data augmentation, we employ random flipping and frame extraction. The initial learning rate is set to $2e^{-4}$ with linear decay. In particular, the pretrained optical flow estimation model [44] is optimized alongside the main model starting at 50K iterations, with a learning rate of 2.5×10^{-5}. The specific network parameter setting will be provided in the *supplementary material*.

Dataset and Metrics We employ the REDS dataset [39] to train our model and comparison methods, which consists of 240 video clips for training and 30 clips for testing. Each clip has 100 frames. As in the mode of degradation mentioned in Section 4, we synthesize degraded video frames online during training to ensure a sufficient diversity of the data. To verify the effectiveness of our proposed method, we test all methods using our proposed Synthetic/Real-world Old Video (SRWOV) Dataset. For the synthetic part, we use the full reference metrics PSNR, SSIM [50], LPIPS [60] and DISTS [9] to quantitatively compare different methods. For real-world part, since the lake of ground truth, we utilize the no-reference quality assessment methods NIQE [38], BRISQUE [37] and CLIPIQA+ [48].

Comparison Methods To validate the effectiveness of the proposed method, we conducted comparative experiments with various approaches. We selected two old photo restoration methods [1, 46], two state-of-the-art old film restoration methods [19, 47], and three state-of-the-art video restoration methods [30, 34, 35]. To ensure a fair comparison, all learning-based methods were trained on the same training set as our proposed method using their official code

5.2. Results

Quantitative Comparisons On the synthetic part of the dataset, we use full-reference metrics to specifically evaluate the quality differences between the restored video frames and the ground truth. As illustrated in Table 1, for pixel-level difference metrics such as PSNR and SSIM, general video restoration methods achieve better performance compared to specialized methods, mainly due to their su-



Figure 4. Visual comparison of the old film restoration on the proposed dataset. The top two rows are the synthetic degradation video, while the bottom row is from the real-world old films. **Zooming in for better comparison.**

Table 1. Quantitative comparisons of the restoration results on synthetic part of the proposed dataset with full-reference metrics. **Bold** text indicates the best performance.

Model	PSNR(dB)↑	SSIM↑	LPIPS↓	DISTS↓
DeOldify[1]	21.9583	0.8312	0.1685	0.1094
OldPhoto[46]	23.7797	0.8247	0.1668	0.0793
DeepRemaster[19]	23.1263	0.8627	0.1826	0.0982
RTN[47]	25.0987	0.8872	0.1025	0.0780
VRT[35]	24.1155	0.8809	0.1249	0.1080
RVRT[34]	24.9226	0.8732	0.1104	0.0842
ShiftNet[30]	26.0208	0.9019	0.1001	0.1160
MambaOFR	**26.9340**	**0.9271**	**0.0538**	**0.0583**

Table 2. Quantitative comparisons of the restoration results on real-world part of the proposed dataset with no-reference metrics. **Bold** text indicates the best performance.

Model	NIQE↓	BRISQUE↓	CLIPIQA+↑
Original	5.7974	43.2733	0.3949
Deoldify[1]	5.3849	39.3407	0.3982
DeepRemaster[19]	5.6132	40.7817	0.4138
OldPhoto[46]	5.6098	41.6991	**0.4482**
RTN[47]	4.8837	28.9296	0.4036
VRT[35]	4.8804	35.0858	0.4079
RVRT[34]	4.8715	33.8577	0.4014
ShiftNet[30]	5.8799	44.5467	0.4027
MambaOFR	**4.8261**	**19.1769**	0.4142

perior baseline. However, in terms of perceptual quality metrics that are closer to human visual preferences, such as LPIPS [60] and DISTS [9], specialized methods exhibit better performance. This is because general methods fall short in eliminating structured defects, which significantly impact visual quality. In contrast, our proposed method shows significant improvements in all metrics. Compared to the best competing methods, our approach achieves an enhancement of over 0.9 dB in PSNR and a 46% improvement in perceptual quality. Additionally, on the real-world part, we use no-reference metrics NIQE [38], BRISQUE [37] and CLIPIQA+ [48] to measure the proximity of the restored video frames to high-quality natural images. As illustrated in Table 2, it can be observed that the restored video frames exhibit higher quality compared to the original versions in terms of metrics. Moreover, our method consistently achieves state-of-the-art performance across multiple metrics.

Qualitative Comparisons To further validate the effectiveness of our method, we provide visual comparisons of the restoration results on both real and synthetic parts of the proposed dataset. As illustrated in Figure 4, we observe that when the original video frames exhibit substantial structural defects, general methods show limited efficacy in restoring these defects, although they can restore photometric degradations. In contrast, our method effectively repairs structural defects. Additionally, due to dynamic degradation awareness of the proposed method, our results more closely resemble the original clean video frames in terms of brightness, sharpness, and other quality aspects. In the real-world frame, it can be observed that the original video frame exhibits black stripe noise at the top. However, many

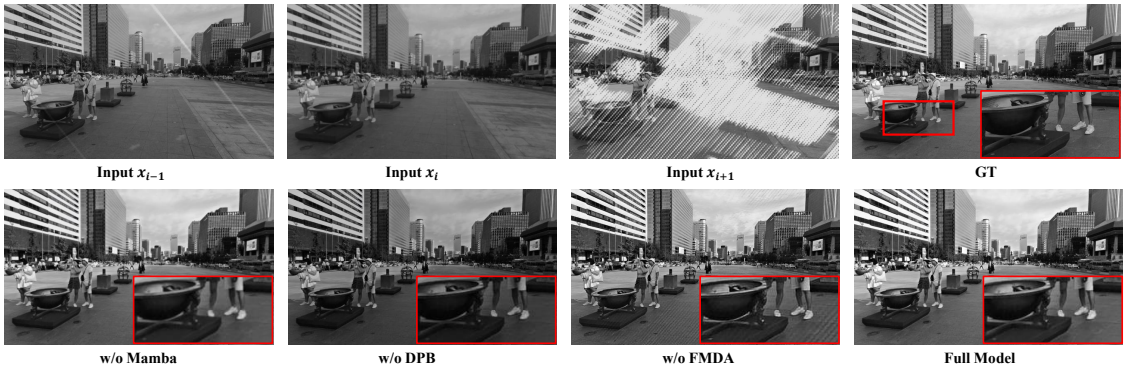


Figure 5. Visual comparison of the ablation study of the proposed MambaOFR. The first three images are partial video frame slices of the input clip, while giving the result and GT for the **input x_i. Zooming in for better comparison.**

restoration methods fail to correctly address this defect in their results. Notably, some general methods even introduce additional defects propagated from previous frames (as seen on the character's face in the image).

5.3. Ablation Study

Mamba Architecture We first consider the improvements of the basic architecture of the model. We replace the mamba block with the Swin-Transformer [36] block. As illustrated in Table 3, performance has decreased across several metrics. This is because long-range modeling and global receptive fields are crucial for video restoration tasks, especially when there are significant occlusions, as local receptive fields struggle to extract meaningful features from the neighborhood. Additionally, as illustrated in Figure 5, the Mamba block effectively enhances detail restoration.

Degradation-aware Prompt Block The degradation-aware prompt block is designed to enhance the network's adaptive capability in dynamic degradation scenarios. We replace it with a standard convolution and attention combination. As illustrated in Table 3, all performance metrics declined. More specifically, as observed in Figure 5, the absence of the DPB makes it challenging for the network to produce optimal low-level features. This is because, unlike tasks with single degradation types, old film restoration needs to handle complex multi-degradation scenarios. Static structures are more susceptible to distribution biases in the training data. Additionally, we verified the impact of increasing the number of parameter groups. Compared to our standard configuration (N=16), fewer parameter groups struggle to fully utilize the potential of the module.

Flow-guided Mask Deformable Alignment An important factor affecting the restoration quality of old films is that, during the aggregation of temporal features, structural defects in certain frames can be propagated to other frames during the feature aggregation process, leading to contamination of the other frames. We replace the proposed flow-guided mask deformable alignment module with the mask-based alignment method from the pervious work [47]. The alignment method proposed, as illustrated in Table 3, similarly improve the performance metrics, while its impact on visual effects is even more pronounced. As show in Figure 5, the restoration result of the input x_i is contaminated by a large-area defect in the input x_{i+1}. Our proposed module can effectively deal with the interference caused by this problem and provide a better restoration effect in the video as a whole.

Table 3. Quantitative comparisons of the restoration results on synthetic part of the proposed dataset with full-reference metrics. **Bold** text indicates the best performance.

Model	PSNR(dB)↑	SSIM↑	LPIPS↓	DISTS↓
w/o Mamba	26.1231	0.8913	0.0876	0.0941
w/o DPB	26.1432	0.9102	0.0743	0.0812
w/o FMDA	26.3467	0.9123	0.0732	0.0741
DPB(N=4)	26.3546	0.9158	0.0729	0.0779
DPB(N=8)	26.5891	0.9211	0.0694	0.0698
MambaOFR	**26.9340**	**0.9271**	**0.0538**	**0.0583**

6. Conclusion and Limitation

In conclusion, the restoration of old films presents unique challenges compared to modern digital video, requiring solutions tailored to analog-specific degradations. Our proposed approach, MambaOFR, establishes a new baseline by integrating degradation-aware prompts that dynamically adjust to the complex and flow-guided mask deformable alignment module that effectively reduces structured defect propagation. Through the introduction of a comprehensive benchmark dataset, we validate that our method achieves state-of-the-art performance.

However, the proposed method still faces certain challenges: (a) In heavily polluted frames, it is difficult to align features using motion estimation from adjacent frames, resulting in poor restoration. (b) In highly degraded videos, recovering fine details from the available features proves challenging, resulting in suboptimal visual outcomes. We hope these challenges can be addressed in future explore.

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